1.7.4 Square root of decimal numbers

To compute the square root of numbers in the decimal form, we simply follow the following steps:

Step 1:

Make even number of decimal places even by affixing a zero on the extreme right of the digit in the decimal part (only if required).

To find $\sqrt{42.25}$

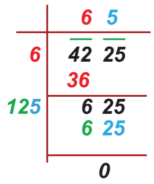
Step 2:

The number has an integral part and a decimal part. In the integral part, mark the bars as done in the case of division method to find the square root of a perfect square number.

We put the bars as $\sqrt{\overline{42}.\overline{25}}$

Step 3:

In the decimal part, mark the bars on every pair of digits beginning with the first decimal place.



Step 4:

Now, calculate the square root by long division method.

Step 5:

Put the decimal point in the square root as soon as the integral part is exhausted. $\therefore \sqrt{42.25} = 6.5$

1.7.5 Square root of product and quotient of numbers

For any two positive numbers a and b. we have

(i) $\sqrt{ab} = \sqrt{a}\times\sqrt{b}$ and (ii) $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ $(b \neq 0)$

Try these

Find the square root of

1. 5.4756 2. 19.36 3. 116.64

Example 1.28

Find the value of $\sqrt{256}$.

Solution:

$$\sqrt{256} = \sqrt{16\times16} = \sqrt{16}\times\sqrt{16} = 4\times4 = 16 \text{ (or) } \sqrt{256} = \sqrt{64\times4} = \sqrt{64}\times\sqrt{4} = 8\times2 = 16.$$

Think

Try to fill in the blanks using $\sqrt{ab} = \sqrt{a}\times\sqrt{b}$.

$\sqrt{36} = 6$	$\sqrt{9}\times\sqrt{4} = 3\times2 = 6$	Is $\sqrt{36} = \sqrt{9}\times\sqrt{4}$?	$\sqrt{81} = ?$	$\sqrt{9}\times\sqrt{9} = \_\_\times\_\_ = \_\_$	Is $\sqrt{81} = \sqrt{9}\times\sqrt{9}$?
$\sqrt{144} = ?$	$\sqrt{9}\times\sqrt{16} = \_\_\times\_\_ = \_\_$	Is $\sqrt{144} = \sqrt{9}\times\sqrt{16}$?	$\sqrt{144} = ?$	$\sqrt{36}\times\sqrt{4} = \_\_\times\_\_ = \_\_$	Is $\sqrt{144} = \sqrt{36}\times\sqrt{4}$?
$\sqrt{100} = ?$	$\sqrt{25}\times\sqrt{4} = \_\_\times\_\_ = \_\_$	Is $\sqrt{100} = \sqrt{25}\times\sqrt{4}$?	$\sqrt{1225} = ?$	$\sqrt{25}\times\sqrt{49} = \_\_\times\_\_ = \_\_$	Is $\sqrt{1225} = \sqrt{25}\times\sqrt{49}$?

Activity

Attempt to prepare a table of square root problems as in the above case to show that if a and b are two perfect square numbers, then $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ $(b \neq 0)$. We can use this idea to compute certain square-root problems easily.